

be maintained. Ferroelectric materials are used in sensors, actuators, memories and energy-harvesting devices. The advantage in using polymers is that these can be liquified using adequate solvents and therefore processed from solution at low cost to form flexible thin films, which are suitable for such electronic devices as capacitors, transistors and diodes. This makes ferroelectric polymers a viable choice for integration with e-textiles.

Scientists at MPI-P, in collaboration with researchers from Johannes Gutenberg University of Mainz and Lodz University of Technology, have developed a method to fabricate ferroelectric nylon thin-film capacitors by dissolving nylon in a mixture of trifluoroacetic acid and acetone, and solidifying it again in vacuum. They were able to realise thin nylon films that are typically only a few hundred nanometres thick, several hundred times thinner than human hair.

A 400-year-old Japanese temple just introduced a robotic priest



Android Kannon, based on Buddhist deity of mercy, preaches sermons at Kodaiji temple in Kyoto, Japan (Credit: Oman Observer)

A 400-year-old Buddhist temple in Kyoto, Japan, is using a one-million-dollar humanoid to reach the younger generation. Kodaiji Temple, in partnership with the robotics team at Osaka University, has unveiled Mindar, a 1.8m (6-feet)-tall Android, made of silicon and aluminium and modelled after Kannon, the Buddhist deity of mercy who preaches what is called Heart Sutra in Japanese, with English and Chinese translations projected on a screen for tourists.

Tensho Goto, chief steward of the temple, told *AFP*, "Artificial intelligence (AI) has developed to such an extent [that] we thought it [was] logical for the Buddha to transform into a robot. Obviously, a machine does not have a soul, but Buddhist faith is not about believing in God. It is about following Buddha's path, so it does not matter whether it is represented by a machine, scrap of iron or tree."

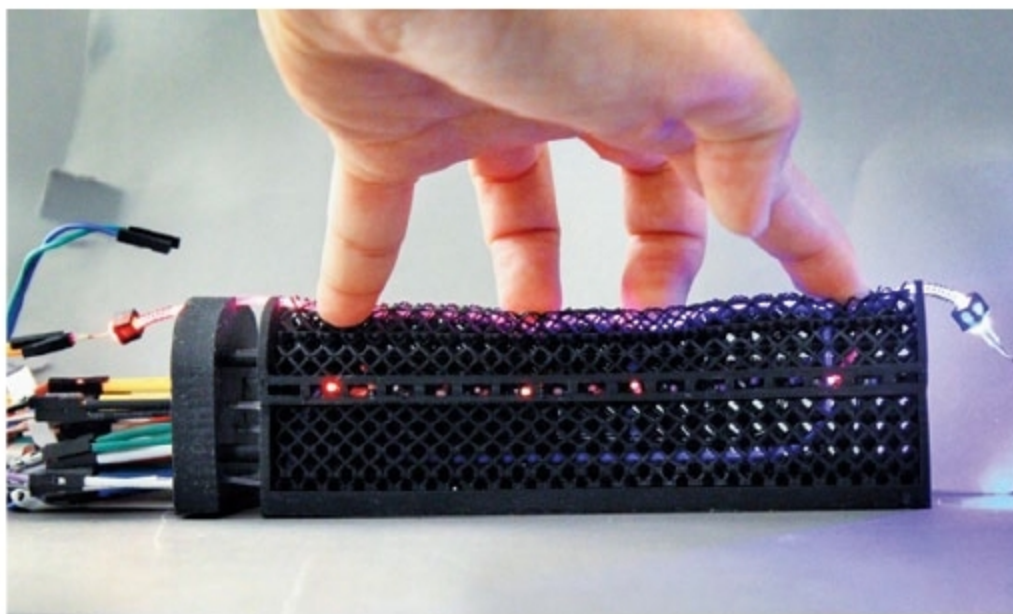
Optical lace that gives robots heightened sensory ability

A new stretchable optical lace can now give soft robots an even softer touch. The synthetic material, developed by Cornell University's PhD student Patricia Xu through Organic Robotics Lab, creates a linked sensory network similar to a biological nervous system, which would enable robots to sense how they interact with their environment and adjust their actions accordingly.

According to lab director Rob Shepherd, associate professor of mechanical and aerospace engineering, humans can feel their environment with their eyes closed, because they have sensors in their fingers that deform when their fingers deform. Robots cannot do that right now.

Earlier, Shepherd's lab had created sensory foams that used optical fibres to detect such deformations. For the optical lace project, Xu used a flexible, porous lattice structure manufactured from 3D-printed polyurethane. She threaded its core with stretchable optical fibres containing more than a dozen mechano-sensors. She then attached an LED light to illuminate the fibre. When she pressed the lattice structure at various points, sensors were able to pinpoint changes in the photon flow.

She says, "When the structure deforms, you have contact between input and output lines, and light jumps into these



Stretchable optical lace gives soft robots an even softer touch (Credit: Cornell University)

output loops in the structure, so you can tell where the contact is happening. The intensity of this determines the intensity of the deformation."

The optical lace will not be used as a skin coating for robots but would be more like the flesh itself. Such robots would be better suited for the healthcare industry, specifically beginning-of-life and end-of-life care, and manufacturing, according to Shepherd.